Jefrey M. Wooldridge. 2002. Econometric Analysis of Cross Section and Panel Data. The MIT Press Cambridge, Massachusetts, London, England.

13.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) In econometrics if an explanatory variable x_j is correlated with the stochastic error u, then the problem of endogeneity arises.

2) The major sources of endogeneity problem in applied econometrics are: Omitted Variables, Measurement Error, and Simultaneity.

3) Measurement error in independent variable x creates an endogeneity bias because it becomes a part of the error term in the regression equation.

   We know that: $X = \tilde{x} - u$ …(1)

   Now putting the value of x in equation $y = \beta x + \in$, we get:

   $y = \beta(\tilde{x} - u) + \in$

   or, $y_i = \beta\tilde{x} + (\in - \beta u)$ … (2)

   Because of positive correlation between $\tilde{x}$ and u, observed in equation (1) the OLS estimator produces bias in $\tilde{\beta}$, which may be negative or positive depending upon the sign of true β in equation (2). If the true β is positive, then bias is negative and if it is negative then bias is positive. The coefficient $\hat{\beta}$ biases towards zero as $0 < \lambda < 1$. This is known as attenuation bias with λ being the attenuation factor. Relative to the true error the residual holds two more sources of variation. First, $\hat{\beta}$ is biased towards zero and the term ($\hat{\beta} - \beta$) does not vanish asymptotically when measurement error is absent. Second, the presence of measurement error in the regressor leads to an increase in error variance.

4) The problem of omitted variables arises when we are unable to include some additional variables due to data unavailability. Suppose $E(y|x,q)$ is the conditional expectation function, linear in parameters and additive in q. However, we shall always estimate $E(y|x)$ for unobserved q. But it has no relationship with $E(y|x,q)$ even if q and x are allowed to be correlated. But we take equation (1) where q is included in the stochastic error u. Thus, x_j is endogenous if q and x_j are correlated.

5) If at least one of the explanatory variables is determined simultaneously along with y then simultaneity arises. Say for example, x_K and u are correlated if x_K is determined partly as a function of y. There are always no sharp distinctions among the three possible forms of endogeneity. We observe more than one source of endogeneity in any given equation.

Check Your Progress 2

1) When an explanatory variable is correlated with the error term, we face the endogeneity problem in a regression model. Consequently, the OLS estimator is not unbiased. As a solution to this problem, we use the instrumental variable method. An instrument should be strongly correlated with the endogenous variable and not correlated with the error term.

2) The IV estimator is consistent due to: the law of large numbers, and $\hat{\beta} = \left(N^{-1}\sum_{i=1}^{N} z_i' x_i\right)^{-1}\left(N^{-1}\sum_{i=1}^{N} z_i' y_i\right) = (Z'X)^{-1} Z'Y$

3) In order to identify β we must give equal importance to the following two conditions: (i) $cov(z_1, u)$, and (ii) $\theta_1 \neq 0$. The major difference between both conditions is that condition (ii) can be tested, whereas condition (i) must be maintained. The covariance in condition (i) involves the unobservable u, and therefore we cannot test anything about $Cov(z_1, u) = 0$. Using t-statistic we can test condition (ii) in its reduced form: $x_K = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_K$

4) The instrumental variable must not be correlated with u so that it is exogenous. See Section 13.3 for details.

Check Your Progress 3

1) The reduced form for x_K is

$$x_K = \delta_0 + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M + r_K$$

2) The linear projection of x_K^* can be written as

$$x_K^* \equiv \delta_0 + \delta_1 x_1 + ... + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M$$

3) The IV estimator using $\hat{x}_i$ as the instruments for x_i can be written as

$$\hat{\beta} = \left(\sum_{i=1}^{N} x_i' x_i\right)^{-1}\left(\sum_{i=1}^{N} \hat{x}_i' y_i\right) = (\hat{X}'X)^{-1} \hat{X}'Y$$

4) See Section 13.5 and answer.

5) There could be M different IV estimators if each z_h has some partial correlation with x_K. In reality there are many more than we can count because any linear combination of $x_1, x_2, . . . , x_{K-1}, z_1, z_2, . . . , z_M$ is uncorrelated with stochastic error u. The two-stage least squares method provides a solution to this problem. should be used. The 2SLS estimator is the most efficient IV estimator.